

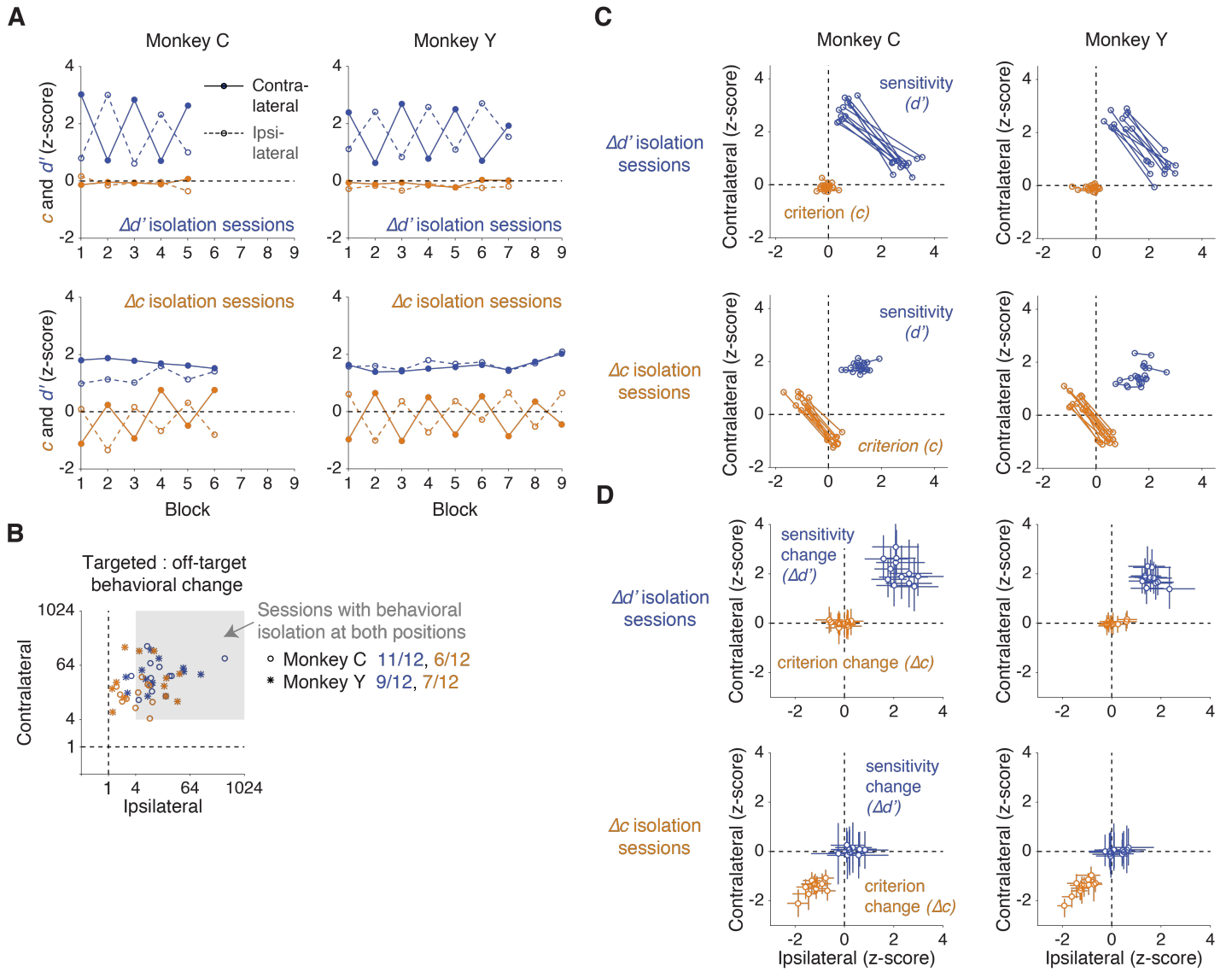


Figure S1. Behavioral performance at the stimulus positions contralateral and ipsilateral to the implanted electrode arrays. Related to Figure 2. (A) During $\Delta d'$ isolation sessions, the subject's d' at the contralateral and ipsilateral locations alternated across blocks between being either high or low while the subject's c at both locations remained stable. Conversely, during Δc isolations, the subject's c at the contralateral and ipsilateral locations changed in counterphase while the subject's d' at both locations remained stable. The counterphase relationship between the two locations indicated that the isolated changes were spatially selective. Data represent average behavior across sessions. Half of the sessions started with the first block attending to the location contralateral to the implanted array, and the other half started with attending to the ipsilateral location. Sessions that started with attending-ipsi were shifted by one block in this plot to align with sessions that started with attending-contra. (B) The ratio of targeted behavioral change to off-target change. In a $\Delta d'$ isolation session, the target:off-target ratio was $\Delta d' / \Delta c$, and in a Δc isolation session, this ratio was $\Delta c / \Delta d'$. A ratio > 4 was considered to indicate a successful isolation. The median ratio was 26:1 at the contralateral location and 8:1 at the ipsilateral location. During 2/48 sessions, not enough errors were obtained at the ipsilateral location to compute c and d' , and isolation was considered unsuccessful at the ipsilateral location. (C) The subject's criterion (c) and sensitivity (d') in each session. Each marker indicates either c (orange) or d' (blue) in one task condition of each session. Markers representing the same session are connected by a line. Error bars were omitted for clarity. (D) The subject's criterion change (Δc) and sensitivity change ($\Delta d'$) in each session.

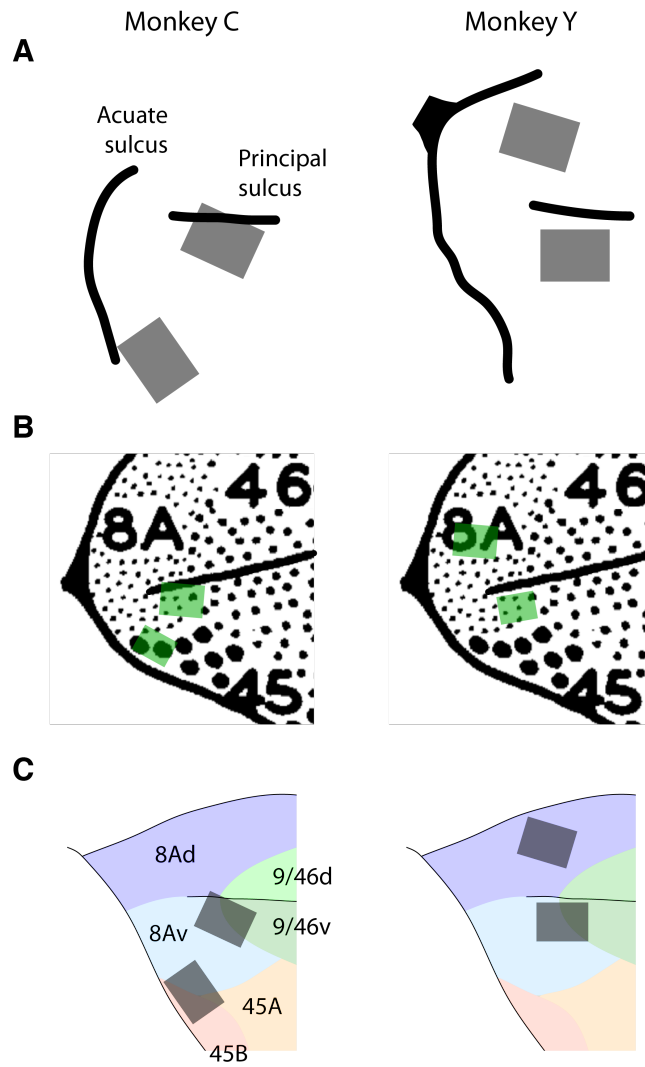


Figure S2. Anatomical locations of microelectrode arrays. Related to STAR Methods. (A) Microelectrode arrays placement relative to frontal sulci. Diagrams were traced from photographs taken during surgery. (B) Diagrams of the arrays superimposed approximately onto the map from (Walker, 1940). For monkey C, the dorsal array extends into areas 8A and 46, and the ventral array extends into areas 8A and 45. For monkey Y, the dorsal array is within area 8A and the ventral array extends into area 8A and 46. (C) Arrays superimposed approximately onto the map from (Petrides and Pandya, 1999). For monkey C, the dorsal array extends into areas 8A and 9/46, and the ventral array extends into areas 8A, 45A, and 45B. For monkey Y, the dorsal array is within area 8A and the ventral array extends into area 8A and 9/46. We refer to these areas as “lateral” prefrontal cortex rather than “dorsolateral” prefrontal cortex because area 45 has been described to be part of “ventrolateral” prefrontal cortex (Petrides and Pandya, 2002).

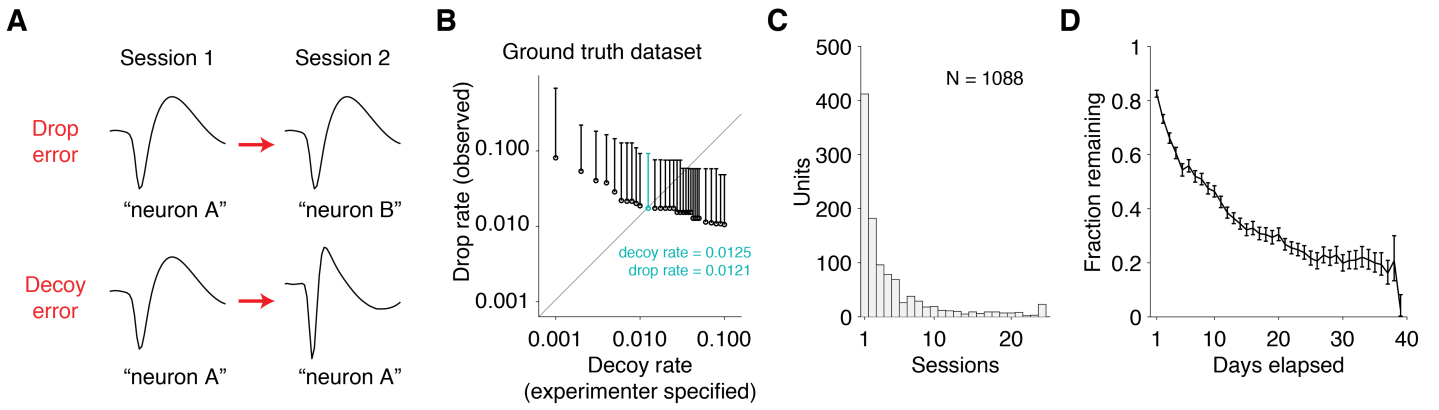


Figure S3. Tracking neurons across sessions. Related to STAR Methods. (A) Two types of error can occur while trying to track the same neuron over multiple sessions. A drop error occurs when the same neuron recorded during two sessions is classified as two different units. A decoy error occurs when two different neurons are classified as the same unit. (B) To assess the accuracy of the tracking algorithm, a ground truth dataset was generated by splitting each recording session into thirds and then comparing the first and last thirds as though they were separate sessions. Each pair of thirds of the same session was processed using different target decoy rates, and the resulting drop rate was measured. The decoy rate of each classifier was specified as the fraction of units from different electrodes (which are almost certainly not the same unit because electrodes were at least 400 μm apart) that would be classified as identical. Each circle indicates the median drop rate across comparisons, and the upper error bar indicates the maximum drop rate across comparisons. The minimum drop rate was 0 for all tested target decoy rates and therefore not plotted. The decoy rate that was most similar to the observed drop rate (cyan) was selected for classifying the full dataset. Note that the drop rate measured here is a lower bound of the drop rate that occurs when tracking units across days. (C) The number of consecutive sessions across which a neuron was tracked. Among the 4,843 units encountered across all days, 1,088 (22% of 4,843) were classified to be unique, and the remaining 3,755 were repeated recordings of the 676 unique units (62% of 1,088) that were recorded for more than one session. (D) The fraction of neurons in an earlier session that was also recorded in a later session, sorted by the intervening number of days between the two sessions. Error bars indicate 95% binomial confidence interval.

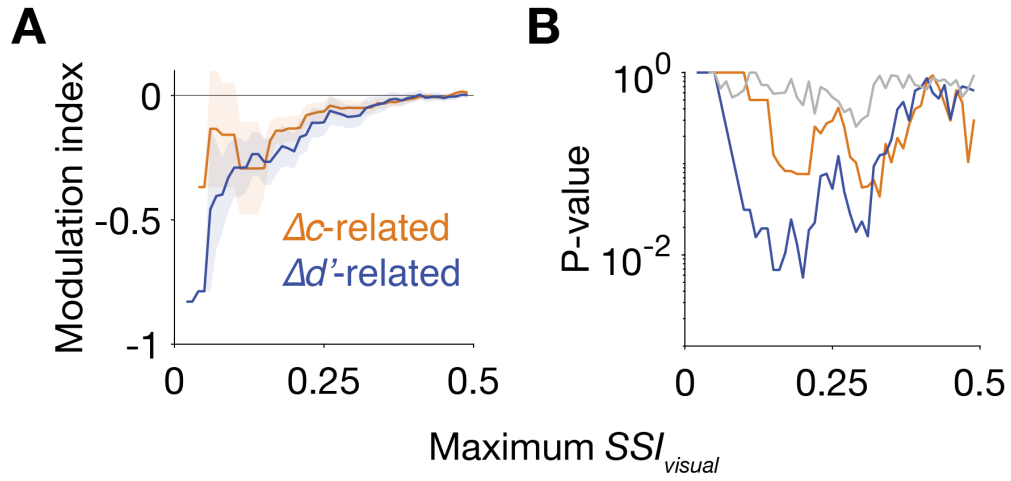


Figure S4. Modulation indices of ipsi-selective visual neurons. Related Figure 4. (A) Modulation index (MI) of firing rates during the period 80-480 ms after sample onset as function of the maximum SSI_{visual} of the units included for analysis. A Δc -related modulation index < 0 indicates stronger firing rates when the animal responded more frequently to ipsilateral location. A $\Delta d'$ -related MI < 0 indicates stronger firing rates when the animal discriminated more accurately at the ipsilateral position. Shading indicates mean $\pm$ SEM across units. Only the 33/48 sessions in which behavior was isolated at both the ipsilateral and contralateral locations were included for this analysis. The sample sizes for maximum $SSI_{visual} = 0.49$ were $N = 278$ (Δc) and $N = 349$ ($\Delta d'$). (B) P-values were computed using the Wilcoxon signed-rank test for a zero median in the distribution of Δc -related modulation indices (orange) or $\Delta d'$ -related indices (blue) and from the Wilcoxon rank-sum test of Δc - and $\Delta d'$ -related indices having the same median (gray).

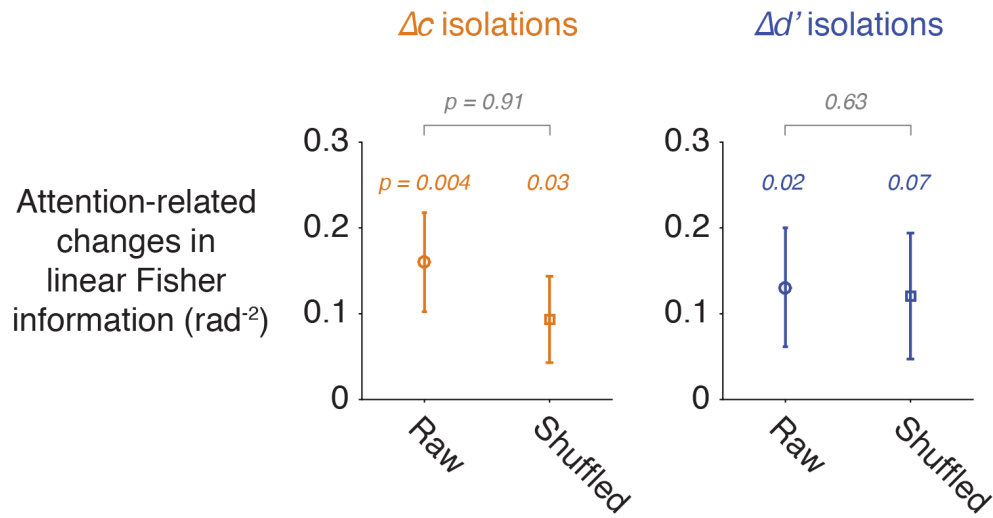


Figure S5. Linear Fisher information, a measure of discriminability of the stimuli by the neural population, increased modestly in association with either decreases in criterion or increases in sensitivity. Related Figure 6. Linear Fisher information is defined as the inverse of the variance of the locally optimal linear decoder of the neural population responses. Higher linear Fisher information indicates better discrimination of the stimuli. Shuffled information quantifies information encoding in the same neural population after correlations between neurons have been removed. Error bars indicate mean $\pm$ SEM across 24 sessions.

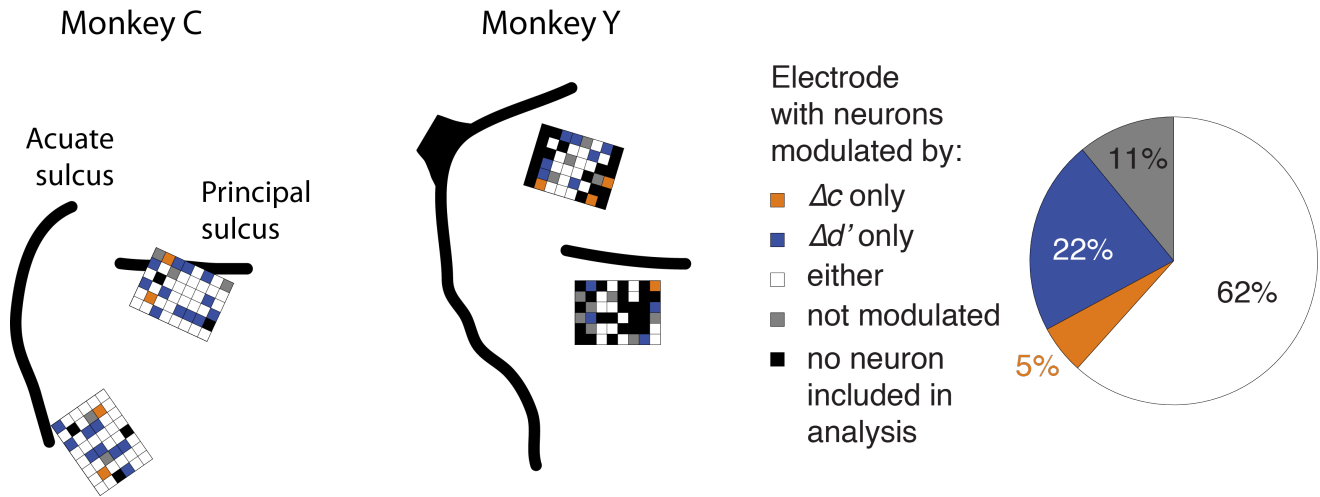


Figure S6. Both Δc -related and $\Delta d'$ -related modulations were observed in all microelectrode arrays and could be observed in the same electrodes (across different days). Related to Figure 8. Each colored square indicates whether the units recorded from an electrode were modulated when the animal changed only its criterion (orange), only its sensitivity (blue), or either (white). Gray indicates that units were not modulated in association with either behavioral change, and black indicates that no unit from that electrode was included for this analysis. A neuron was included for this analysis if its firing rates were modulated during the sample period of the attention task relative to the fixation period at $p = 0.01$ (Wilcoxon rank-sum test) and also have a trial-averaged firing rate of at least 5 Hz during any task period. Sixty-five percent of units recorded was included. Neurons were then sorted according to their electrode and the type of isolation session when they were recorded. For each group of neurons from the same electrode and same isolation, an ANOVA with factors *neuron* and *attention* was performed. The *attention* factor indicates whether firing rate was modulated in relation to either Δc or $\Delta d'$. An electrode was considered to be modulated if the main effect of *attention* had a p-value less than 0.01. The pie chart includes only significantly modulated electrodes.

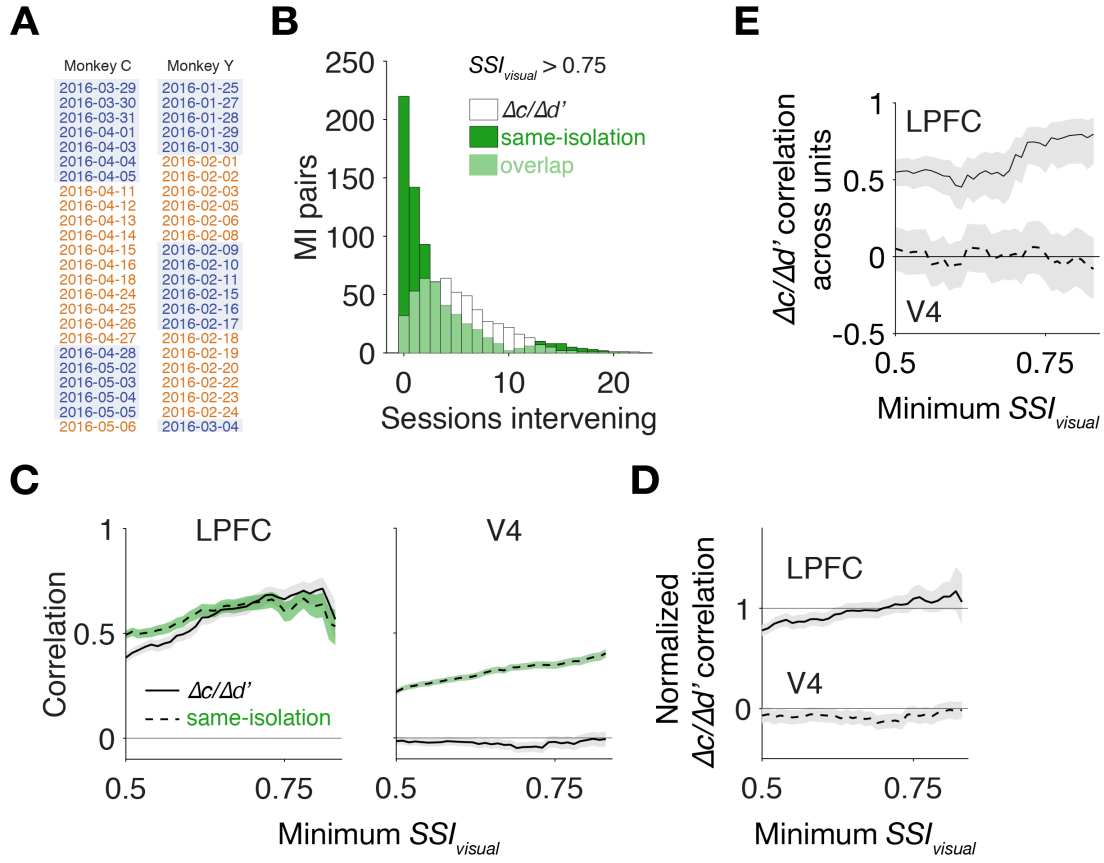


Figure S7. Additional analyses related to Figure 8. (A) The dates of the 24 Δc -isolations (orange) and $\Delta d'$ -isolations (blue) during which neurons from LPFC were recorded. Because the type of isolation switched only after at least five successful sessions, $\Delta c/\Delta d'$ sessions pairs were on average spaced further apart in time. (B) The number of intervening sessions between each pair of MI. Each MI pair was recorded from the same unit, and all pairs of MI's were pooled across all units. To match the distributions, for each bin, a number of MI pairs was randomly selected from each of the $\Delta c/\Delta d'$ distribution (white) and same-isolation distribution (green), and this number was equal to the overlap between the two distributions (light green) in that bin. Subsets of MI pairs were pooled across bins to calculate the correlation coefficient. Random subsampling was repeated 1,000 times to provide 1,000 correlation coefficients and confidence intervals. The median correlation coefficient and its associated confidence interval were shown in **Figures 8C-D**. (C-D) The same analyses in **Figure 8C-D** performed *without* distribution-matching, showing that it did not qualitatively change the results. (E) An alternative analysis that computed correlations across units rather than pairs of MIs similarly found a difference in $\Delta c/\Delta d'$ correlations between LPFC and V4. In this analysis, a single Δc -related MI and a single $\Delta d'$ -related MI was computed for each unit by averaging across MIs from different sessions in which that unit was recorded. Shading indicates 95% confidence interval.

Model	$\Delta d'$ -related modulations		Δc -related modulations	
	Variance explained (%)	BIC	Variance explained (%)	BIC
$MI \sim b_0$	<1	251.48	<1	100.68
$MI \sim b_0 * \text{visual}$	14.6	35.23	9	18.39
$MI \sim b_0 * \text{delay}$	<1	256.23	<1	110.3
$MI \sim b_0 * \text{presac}$	<1	256.69	<1	129.05
$MI \sim b_0 * \text{postsac}$	<1	321.57	<1	162.22
$MI \sim b_0 + b_1 * \text{visual}$	15.4	29.13	10.8	8.25
$MI \sim b_0 + b_1 * \text{delay}$	3.8	206.04	4.1	71.39
$MI \sim b_0 * \text{visual} + b_1 * \text{delay}$	15.4	28.36	10.4	11.52
$MI \sim b_0 * \text{visual} + b_1 * \text{delay} + b_2 * \text{visual} : \text{delay}$	16.2	23.62	11.8	4.5
$MI \sim b_0 + b_1 * \text{presac}$	3.8	205.11	2.4	86.65
$MI \sim b_0 * \text{visual} + b_1 * \text{presac}$	16.3	13.91	10.6	10.19
$MI \sim b_0 * \text{visual} + b_1 * \text{presac} + b_2 * \text{visual} : \text{presac}$	16.9	12.1	11.7	5.59
$MI \sim b_0 * \text{delay} + b_1 * \text{presac}$	1	244.79	<1	113.33
$MI \sim b_0 * \text{delay} + b_1 * \text{presac} + b_2 * \text{delay} : \text{presac}$	4.6	200.65	4.1	78.15
$MI \sim b_0 + b_1 * \text{postsac}$	<1	251.44	<1	107.42
$MI \sim b_0 * \text{visual} + b_1 * \text{postsac}$	15.7	23.29	9.1	24.41
$MI \sim b_0 * \text{visual} + b_1 * \text{postsac} + b_2 * \text{visual} : \text{postsac}$	16.5	18.41	10.7	15.71
$MI \sim b_0 * \text{delay} + b_1 * \text{postsac}$	<1	250.24	<1	116.59
$MI \sim b_0 * \text{delay} + b_1 * \text{postsac} + b_2 * \text{delay} : \text{postsac}$	4.5	202.26	4.2	76.74
$MI \sim b_0 * \text{presac} + b_1 * \text{postsac}$	<1	257.79	<1	135.77
$MI \sim b_0 * \text{presac} + b_1 * \text{postsac} + b_2 * \text{presac} : \text{postsac}$	4.3	206.21	2.3	94.21
$MI \sim b_0 + b_1 * \text{visual} + b_2 * \text{delay}$	16.2	23.9	11.9	3.4
$MI \sim b_0 + b_1 * \text{visual} + b_2 * \text{delay} + b_3 * \text{visual} : \text{delay}$	16.2	30.8	12.1	8.91
$MI \sim b_0 + b_1 * \text{visual} + b_2 * \text{presac}$	17	10.74	11.9	3.43
$MI \sim b_0 + b_1 * \text{visual} + b_2 * \text{presac} + b_3 * \text{visual} : \text{presac}$	17.3	12.23	13	0
$MI \sim b_0 + b_1 * \text{delay} + b_2 * \text{presac}$	4.8	198.35	4.3	76.29
$MI \sim b_0 + b_1 * \text{delay} + b_2 * \text{presac} + b_3 * \text{delay} : \text{presac}$	5	202.78	4.4	81.42
$MI \sim b_0 * \text{visual} + b_1 * \text{delay} + b_2 * \text{presac}$	16.4	20.29	10.9	13.67
$MI \sim b_0 * \text{visual} + b_1 * \text{delay} + b_2 * \text{presac} + b_3 * \text{visual} : \text{delay} + b_4 * \text{visual} : \text{delay} : \text{presac}$	17.4	17.46	12.9	7.75
$MI \sim b_0 * \text{visual} + b_1 * \text{delay} + b_2 * \text{presac} + b_3 * \text{visual} : \text{delay} + b_4 * \text{visual} : \text{presac}$	17.6	15.11	12.5	11.63
$MI \sim b_0 + b_1 * \text{visual} + b_2 * \text{postsac}$	16.5	17.5	10.8	14.73
$MI \sim b_0 + b_1 * \text{visual} + b_2 * \text{postsac} + b_3 * \text{visual} : \text{postsac}$	16.6	24.02	10.9	20.93
$MI \sim b_0 + b_1 * \text{delay} + b_2 * \text{postsac}$	4.7	199.8	4.1	78.12
$MI \sim b_0 + b_1 * \text{delay} + b_2 * \text{postsac} + b_3 * \text{delay} : \text{postsac}$	4.9	204.7	4.3	82.99
$MI \sim b_0 * \text{visual} + b_1 * \text{delay} + b_2 * \text{postsac}$	16.8	12.49	10.5	17.62
$MI \sim b_0 * \text{visual} + b_1 * \text{delay} + b_2 * \text{postsac} + b_3 * \text{delay} : \text{postsac}$	17.5	8.7	12.1	8.67
$MI \sim b_0 * \text{visual} + b_1 * \text{delay} + b_2 * \text{postsac} + b_3 * \text{visual} : \text{delay} : \text{postsac}$	17.6	7.77	12	10.1
$MI \sim b_0 + b_1 * \text{presac} + b_2 * \text{postsac}$	4.3	205.27	2.4	93.12
$MI \sim b_0 + b_1 * \text{presac} + b_2 * \text{postsac} + b_3 * \text{presac} : \text{postsac}$	4.3	212.19	2.4	99.57
$MI \sim b_0 * \text{visual} + b_1 * \text{presac} + b_2 * \text{postsac}$	17.4	2.97	10.6	16.92
$MI \sim b_0 * \text{visual} + b_1 * \text{presac} + b_2 * \text{postsac} + b_3 * \text{presac} : \text{postsac}$	18	0	11.9	10.26
$MI \sim b_0 * \text{delay} + b_1 * \text{presac} + b_2 * \text{postsac}$	1.8	241.57	<1	120
$MI \sim b_0 * \text{delay} + b_1 * \text{presac} + b_2 * \text{postsac} + b_3 * \text{delay} : \text{postsac}$	5.4	196.65	4.4	81.47
$MI \sim b_0 * \text{delay} + b_1 * \text{presac} + b_2 * \text{postsac} + b_3 * \text{presac} : \text{postsac}$	5.5	195.14	4.2	83.51
$MI \sim b_0 + b_1 * \text{visual} + b_2 * \text{delay} + b_3 * \text{presac}$	17	17.24	12.3	6.97
$MI \sim b_0 + b_1 * \text{visual} + b_2 * \text{delay} + b_3 * \text{presac} + b_4 * \text{visual} : \text{presac}$	17.3	18.56	13.3	3.47
$MI \sim b_0 + b_1 * \text{visual} + b_2 * \text{delay} + b_3 * \text{postsac}$	17.5	8.58	12	9.91
$MI \sim b_0 + b_1 * \text{visual} + b_2 * \text{delay} + b_3 * \text{postsac} + b_4 * \text{delay} : \text{postsac}$	17.5	15.8	12.3	13.57
$MI \sim b_0 + b_1 * \text{visual} + b_2 * \text{delay} + b_3 * \text{postsac} + b_4 * \text{visual} : \text{delay}$	17.6	14.66	12.1	15.49
$MI \sim b_0 + b_1 * \text{visual} + b_2 * \text{presac} + b_3 * \text{postsac}$	18	0.02	11.9	10.2
$MI \sim b_0 + b_1 * \text{visual} + b_2 * \text{presac} + b_3 * \text{postsac} + b_4 * \text{visual} : \text{presac}$	18.3	2.67	13	6.65
$MI \sim b_0 + b_1 * \text{delay} + b_2 * \text{presac} + b_3 * \text{postsac}$	5.6	194.63	4.3	83.06
$MI \sim b_0 + b_1 * \text{delay} + b_2 * \text{presac} + b_3 * \text{postsac} + b_4 * \text{delay} : \text{postsac}$	5.7	200.3	4.5	87.41
$MI \sim b_0 + b_1 * \text{delay} + b_2 * \text{presac} + b_3 * \text{postsac} + b_4 * \text{delay} : \text{presac} : \text{postsac}$	5.7	199.73	4.3	89.78
$MI \sim b_0 * \text{visual} + b_1 * \text{delay} + b_2 * \text{presac} + b_3 * \text{postsac}$	17.6	7.8	10.9	20.29
$MI \sim b_0 * \text{visual} + b_1 * \text{delay} + b_2 * \text{presac} + b_3 * \text{postsac} + b_4 * \text{visual} : \text{presac} + b_5 * \text{delay} : \text{postsac}$	18.3	10.26	13.5	7.74
$MI \sim b_0 * \text{visual} + b_1 * \text{delay} + b_2 * \text{presac} + b_3 * \text{postsac} + b_4 * \text{visual} : \text{delay} + b_5 * \text{visual} : \text{presac}$	18.8	0.86	12.5	18.38
$MI \sim b_0 + b_1 * \text{visual} + b_2 * \text{delay} + b_3 * \text{presac} + b_4 * \text{postsac}$	18.2	5.08	12.3	13.71
$MI \sim b_0 + b_1 * \text{visual} + b_2 * \text{delay} + b_3 * \text{presac} + b_4 * \text{postsac} + b_5 * \text{visual} : \text{presac}$	18.4	7.52	13.3	10.21
$MI \sim b_0 + b_1 * \text{visual} + b_2 * \text{delay} + b_3 * \text{presac} + b_4 * \text{postsac} + b_5 * \text{visual} : \text{delay} + b_6 * \text{visual} : \text{presac}$	19	5.86	13.4	15.69

Table S1. Linear models of firing rate modulation indices using SSI's as predictor variables. Related to Figure 5. Modulation indices of firing rate (MI) were computed from spike counts during the sample period in the attention task. The optimal model was selected according to the lowest Bayesian Information Criterion (BIC). BIC values are displayed relative to the minimum across models. To simplify presentation, for each set of models that have the same variables but different combinations of interaction terms, we show only the model that has the lowest BIC for each type of modulation. The number of models considered was 4,257.